Bianca Bochner

Penlink מבית InsAlghts Academy בשיתוף Google and Reichman AI Tech School

Script SQL: visualizations

רשת המרכולים "שפע יששכר"

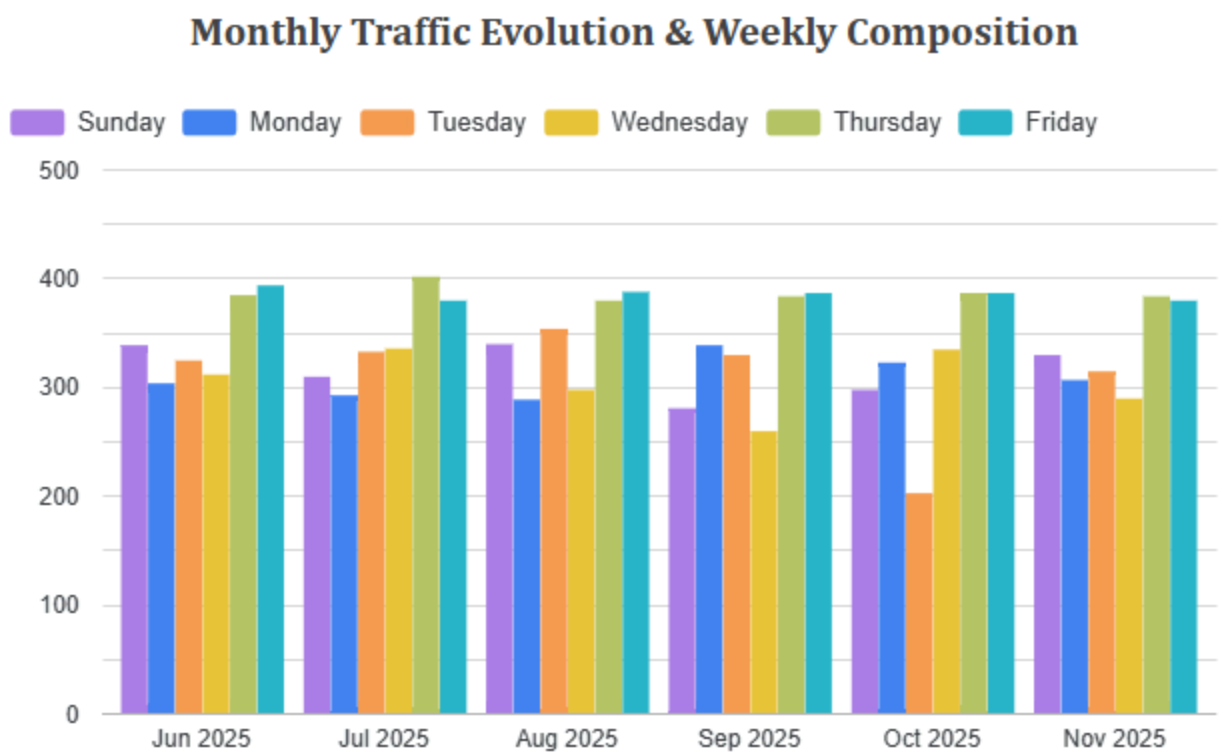
The Shefa Be-Paa Initiative: Orchestrating Retail Efficiency through Advanced Data Modeling

➔ Checking distinct roles

Row	role	total_devices_per_role
1	butcher	4
2	not_paying	276
3	one_time_customer	409
4	security_guy	4
5	cashier	15
6	delivery_guy	8
7	manager	1
8	general_worker	10
9	repeat_customer	352
10	senior_general_worker	1

Row	area
1	CASH_REGISTERS
2	BUTCHERY
3	HEAD_OFFICE
4	SUPERMARKET
5	PARKING
6	WAREHOUSE

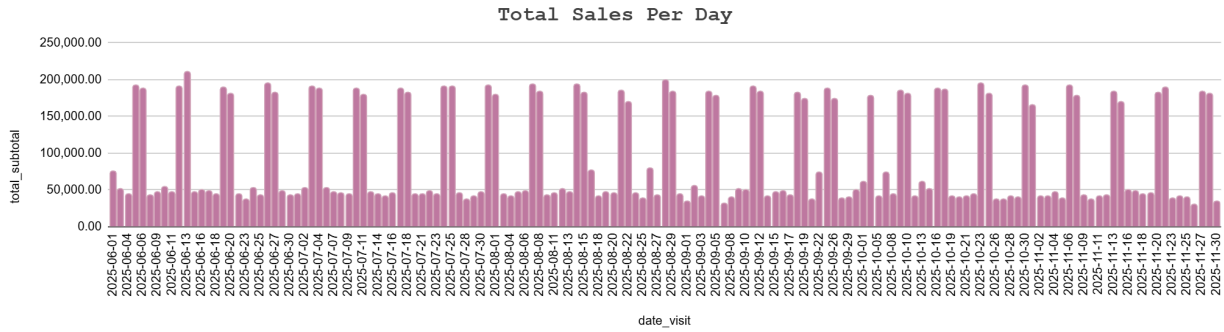
➔ Clients per month and day of week



This chart provides a detailed view of store traffic trends from June to November 2025. Here is a brief analysis of the findings:

- Weekly Peak Period: The chart reveals a consistent surge in traffic toward the end of the week, with Thursday (Green) and Friday (Cyan) reaching the highest customer counts across all observed months.
- Stability and Growth: Overall traffic remains relatively stable throughout the period, maintaining a baseline of approximately 300 to 400 customers per day.
- Tuesday Volatility: While most days show steady performance, there is a significant and notable dip in Tuesday (Orange) traffic during October 2025 compared to the preceding and following months.
- Operational Insight: The dominance of end-of-week traffic suggests that staffing and operational support should be heavily prioritized on Thursdays and Fridays to accommodate the consistently higher volume.

➔ Daily sales



➔ Payment Methods

Row	payment_method	total_transactions	percentage
1	credit_card	15162	43.83
2	debit_card	14989	43.33
3	cash	2232	6.45
4	mobile_pay	2206	6.38

➔ Customer Behavior & App Engagement

role	total_activations	overall_total	percentage_of_to...
repeat_customer	538944	893044	60.35
not_paying	65740	893044	7.36
one_time_customer	17166	893044	1.92

➔ App Penetration in Sales

total_sales	app_customers	percentage_app_customers	no_app_customers	percentage_no_app_customers
34589	30993	89.6	3596	10.4

➔ Identifying and treating outliers in location transmission accuracy level

Row	total_signal	filtered_accuracy_m	percentage_filtered_accuracy
1	1308624	893044	68.24

➔ Identify all distinct role and area combinations in the geolocation data

Row	role	area
1	butcher	BUTCHERY
2	cashier	CASH_REGISTERS
3	delivery_guy	WAREHOUSE
4	general_worker	SUPERMARKET
5	manager	HEAD_OFFICE
6	manager	SUPERMARKET
7	not_paying	SUPERMARKET
8	not_paying	PARKING
9	one_time_customer	PARKING
10	one_time_customer	SUPERMARKET
11	one_time_customer	CASH_REGISTERS
12	repeat_customer	SUPERMARKET
13	repeat_customer	CASH_REGISTERS
14	repeat_customer	PARKING
15	security_guy	PARKING
16	senior_general_worker	SUPERMARKET

➔ Accuracy outlier analysis by role

Row	role	total_outliers	total_filtered_recor...	proportional_outlier_rate	outlier_rate_perc...
1	security_guy	43533	49109	0.88645665763912929	88.65%
2	repeat_customer	309326	848270	0.36465512160043384	36.47%
3	butcher	18800	60458	0.3109596744847663	31.10%
4	manager	8297	31347	0.26468242575047052	26.47%
5	not_paying	15610	81350	0.19188690842040565	19.19%
6	cashier	17571	91581	0.19186294100304649	19.19%
7	one_time_customer	2443	19609	0.12458564944668264	12.46%
8	delivery_guy	0	1115	0.0	0.00%
9	senior_general_work...	0	42644	0.0	0.00%
10	general_worker	0	83141	0.0	0.00%

➔ geolocation accuracy outlier rate by area

area	total_signals	total_outliers	outlier_rate	outlier_rate_percenta...
PARKING	127639	69928	0.5479	54.79
BUTCHERY	60458	18800	0.311	31.1
SUPERMARKET	968850	291036	0.3004	30.04
HEAD_OFFICE	29408	7784	0.2647	26.47
CASH_REGISTERS	121154	28032	0.2314	23.14
WAREHOUSE	1115	0	0.0	0.0

➔ Validation of dwell_minutes (In-Store Consistency Check)

avg_dwell_ratio	avg_dwell_ratio_pct	visits_analyzed
0.88243659736432334	88.24	30896

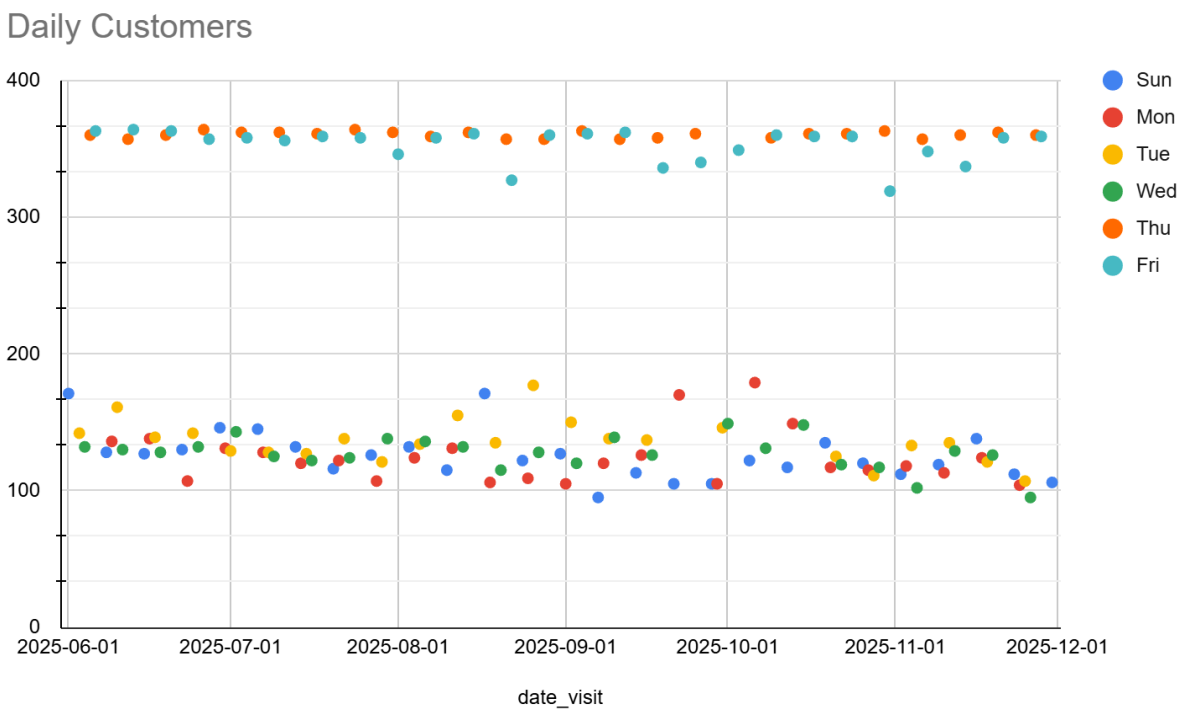
➔ App capture rate - In-Store only (Official Metric)

avg_capture_rate	avg_capture_rate...	visits_analyzed
0.318842618151...	31.88426181512...	34877

➔ Quantifying the impact of parking on dwell time

avg_parking_minutes_added	avg_parking_pct_of_total_dwell	avg_parking_pct_vs_in_store	visits_analyzed
4.8754488590293139	10.480767975217422	19.632758695000764	34532

➔ Daily frequency of customers



➔ Customer Traffic Density & Behavioral Profiling

Jun 1, 2025 - Nov 30, 2025

Customer Traffic Density & Behavioral Profiling

HOUR	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
8AM	-	-	-	-	-	604.31
9AM	287.96	280.48	282.35	267.32	-	1,419.5
10AM	259.33	245.56	273.17	231.12	779.92	984.19
11AM	289.22	284	284.96	272.04	1,053.64	736.73
12PM	273.89	248.28	252.83	229.44	985.68	1,401.65
1PM	295.3	266.68	318.96	284.08	1,048.2	1,613.54
2PM	250.78	224.28	260.91	253.76	195.52	1,004.62
3PM	10.44	15.25	13.8	11.33	2.38	-
4PM	-	-	-	-	791.96	-
5PM	274.52	276.72	297.74	285.88	1,043.16	-
6PM	246.63	246.92	269.43	275.12	972.96	-
7PM	13.16	11.16	11.22	12.4	1,063.92	-
8PM	-	-	-	-	183.32	-
9PM	-	-	-	-	2	-

➔ Financial customer segmentation: ticket × frequency

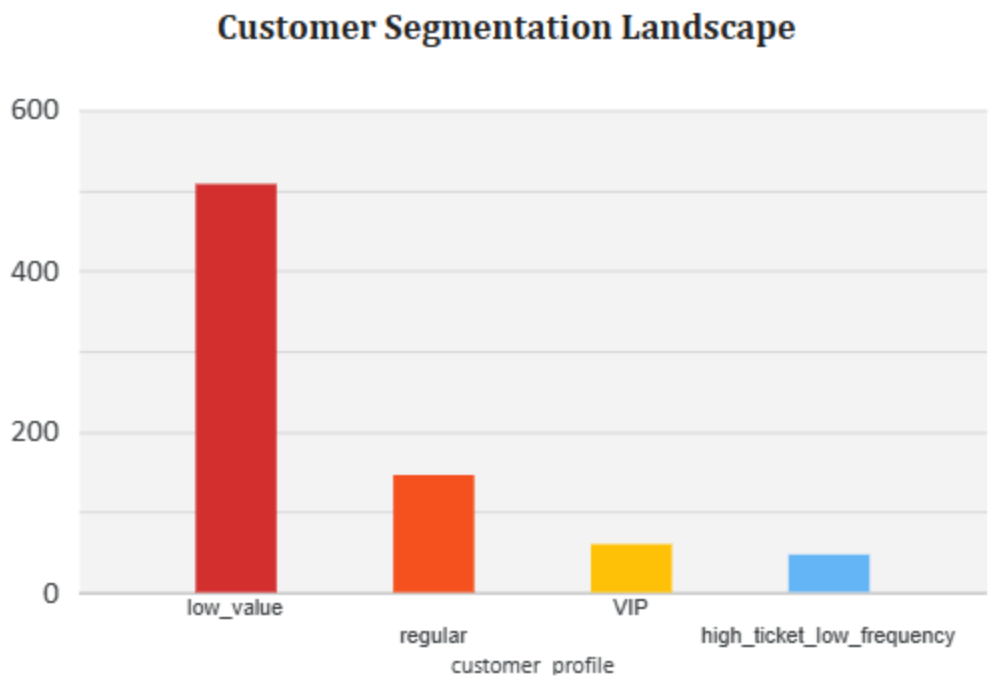


Figure: Distribution of Strategic Customer Segments

Legend:

- VIP: High Spend (>=P75) & High Frequency (>=P75).
- Regular: Median behavior across both dimensions (P25 to P75).
- Low Value: Bottom tier performance (<=P25) in both metrics.
- Specialized Segments: High-Ticket/Low-Frequency (Big basket/Rare visits) and Low-Ticket/High-Frequency (Small basket/Loyal).

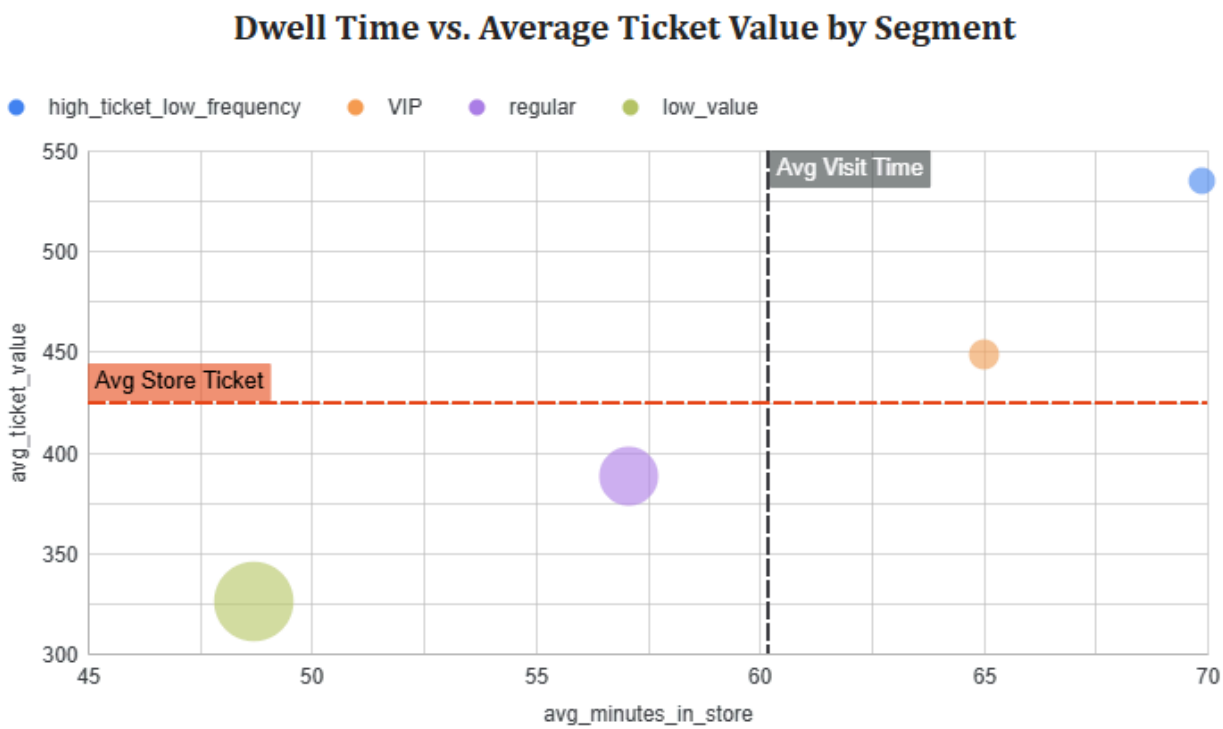
➔ Revenue Impact Analysis by Segment

customer_profile	total_customers	group_revenue	revenue_share_pct	customer_share_pct	avg_ltv
low_value	508	8129977.81	61.71	66.75	16003.89
regular	146	2650899.52	20.12	19.19	18156.85
VIP	60	2362987.88	17.94	7.88	39383.13
high_ticket_low_frequency	47	29577.61	0.22	6.18	629.31

➔ Dwell Time and Efficiency Analysis by Segment

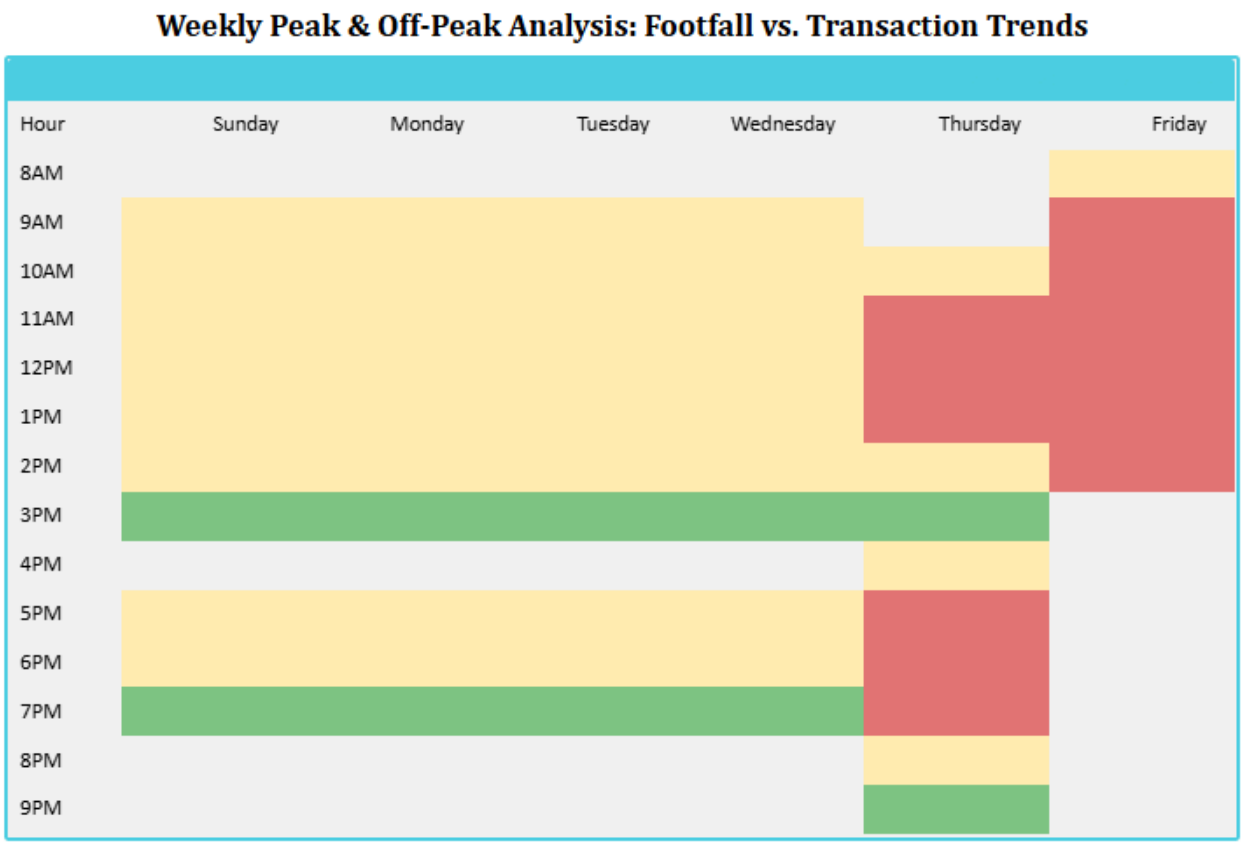
customer_profile	total_customers	avg_minutes_in_store	avg_ticket_value	revenue_per_minute
high_ticket_low_frequency	47	69.9	534.77	7.65
VIP	60	65.0	448.7	6.9
regular	232	57.1	387.67	6.78
low_value	422	48.7	325.83	6.69

Customer Monetization Efficiency Mapping



This Scatter Analysis maps the 'Monetization Velocity' of the store. By cross-referencing Dwell Time with Ticket Value, the analysis identifies that VIPs provide the highest ROI per minute spent on the floor. The quadrants highlight segments that stay too long with low returns, signaling a need for self-service optimization.

➔ Combined Customers × Transactions: Percentile-Based Composite Activity Index



This heatmap visualizes a Composite Store Activity Index, which balances physical customer presence (60% weight) with real-time transaction volume (40% weight). By normalizing these metrics into percentiles, the chart identifies specific "operational states" regardless of raw

numbers. It allows management to see at a glance when the store is under the most pressure versus when it is under-utilized.

➔ Staffing analysis with peak / dead hour classification (p75 Demand)

Staffing analysis with peak / dead hour classification

HOUR	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
8AM	-	-	-	-	-	
9AM					-	
10AM						
11AM						
12PM						
1PM						
2PM						
3PM						-
4PM	-	-	-	-		-
5PM						-
6PM						-
7PM						-
8PM	-	-	-	-		-

Interpretation of Results

- *Peak_hour*: requires operational reinforcement (staffing or self-checkout support)
- *Dead_hour*: opportunity for cost optimization and automation
- *Normal_hour*: balanced operation

➔ Integrated staffing × demand × self-checkout decision KPI

Self-Checkout Operational Decision Matrix

HOUR	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
8AM						
9AM						
10AM						
11AM						
12PM						
1PM						
2PM						
3PM						
4PM						
5PM						
6PM						

Operational Support Cost Optimization

The visualization serves as a dynamic staffing guide. It identifies the critical hours where Self-Checkout acts as a safety net for high demand (🟧Operational Support) and when it serves as the primary service point to reduce labor overhead during low-traffic periods (🟦Cost Optimization).

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